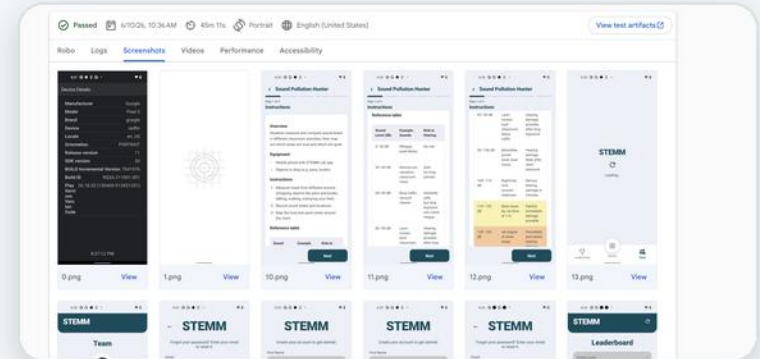




What it is

A mobile app where student teams complete hands-on science, engineering, and health challenges using real device sensors and evidence capture.



Tech stack

- Expo React Native
- TypeScript
- Firebase Auth + Firestore
- SQLite local session capture

Learning loop

- Predict
- Record real data
- Write up results
- Discuss and earn points

Repo evidence

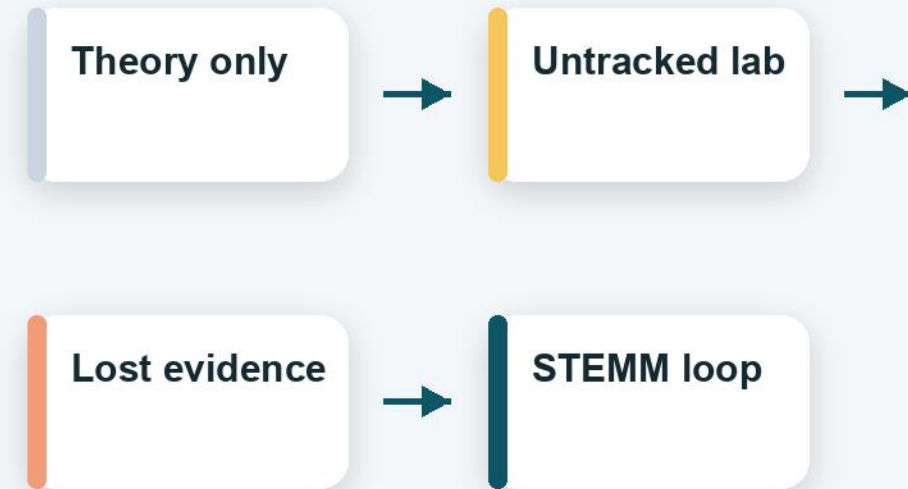
- 7 activity routes
- Jest: 108 tests passing
- Firebase Test Lab Robo run passed

The Problem

Why STEMM is needed

2/8

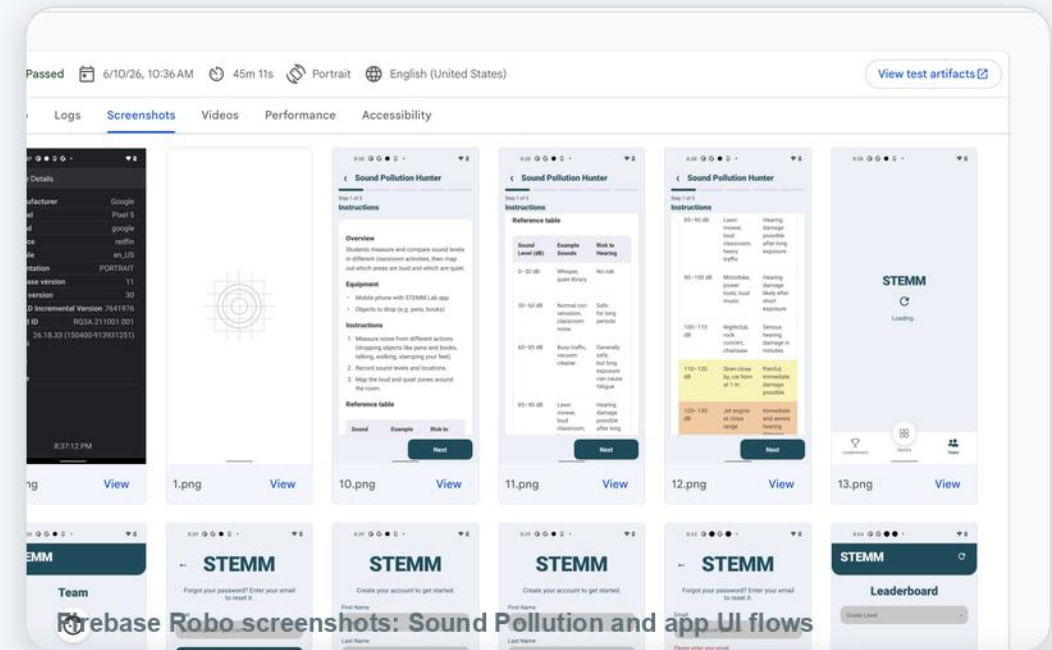
- STEM learning can stay theoretical without enough practical feedback.
- Classroom labs are hard to run consistently across teams.
- Manual tracking makes predictions, evidence, reflections, and scoring easy to lose.
- Teachers need team progress and results in one place.



STEMM closes the loop: guided workflow + device data + saved evidence + team leaderboard.



- Same 5-step flow across every activity.
- Recorder step validates required evidence before Next.
- Screens save prediction, data points, video evidence, rating, and GPS.
- Hardware examples: microphone, motion sensor, video picker, timers.



Teams, Activities & Leaderboard

Auth, invite codes, scoring, and ranking

4/8

Teams

Create or join with a 6-character invite code. Codes avoid ambiguous I/O/0/1 characters.

Activities

4 engineering challenges + 3 health and medical science games.

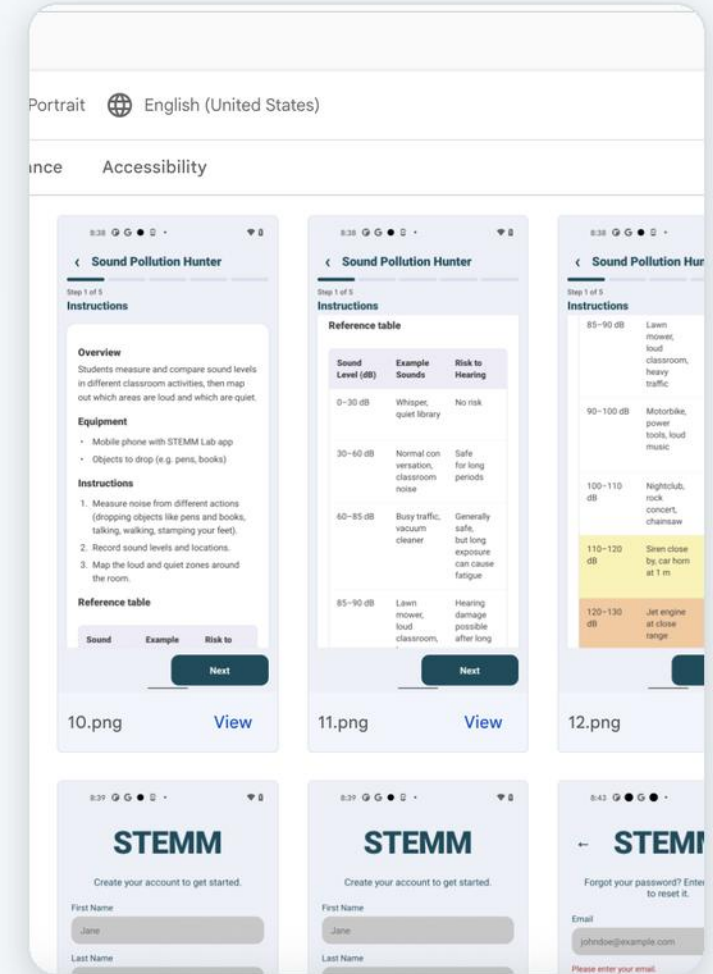
Leaderboard

Firestore teams ranked by score, filterable by grade level, with pull-to-refresh.

Point scheme

Every activity completion awards 100 team points. Duplicate scoring is blocked by point_awards/{teamId}_{activityId}.

Parachute, Sound, Fan, Earthquake, Human Performance, Reaction Board, Breathing Pace



Automated result

9 test suites passed
108 tests passed
0 failed
0 snapshots

Coverage layers

Unit: points, CRUD, notifications,
discriminant, background tasks
Integration: Firebase, errors, theme
E2E/config: app.json contract

Device QA

Firebase Test Lab Robo test passed
on Pixel 5, API 30, portrait, en-US.

```
at callCircusTest (C:\Users\denni\Documents\CodingProjects\STEMM\node_modules\jest-circus\build\run.js:116:40)
at runTest (C:\Users\denni\Documents\CodingProjects\STEMM\node_modules\jest-circus\build\run.js:252:11)
at runTestsForDescribeBlock (C:\Users\denni\Documents\CodingProjects\STEMM\node_modules\jest-circus\build\run.js:126:9)
at runTestsForDescribeBlock (C:\Users\denni\Documents\CodingProjects\STEMM\node_modules\jest-circus\build\run.js:126:9)
at run (C:\Users\denni\Documents\CodingProjects\STEMM\node_modules\jest-circus\build\run.js:121:3)
at runAndTransformResultsToTestFormat (C:\Users\denni\Documents\CodingProjects\STEMM\node_modules\jest-circus\build\legacy-code-todo-rewrite\jestAdapterInit.js:122:21)
at jestAdapter (C:\Users\denni\Documents\CodingProjects\STEMM\node_modules\jest-circus\build\legacy-code-todo-rewrite\jestAdapter.js:79:19)
at runTestInternal (C:\Users\denni\Documents\CodingProjects\STEMM\node_modules\jest-runner\build\runTest.js:167:16)
at runTest (C:\Users\denni\Documents\CodingProjects\STEMM\node_modules\jest-runner\build\runTest.js:144:14)

19 |   return BackgroundTask.Result.Success;
20 | } catch (error) {
> 21 |   console.error("Background sync failed:", error);
    |   ^
22 |   return BackgroundTask.Result.Failed;
23 | }
24 | });

at Object.error (lib/background-tasks.ts:21:13)
at Generator.throw (anonymous)
at asyncGeneratorStep (node_modules\@babel/runtime\helpers\asyncGenerator.js:3:17)
at _throw (node_modules\@babel/runtime\helpers\asyncGenerator.js:20:9)

console.error
Background sync failed: Error: permission-denied
    at Object.<anonymous> (C:\Users\denni\Documents\CodingProjects\STEMM\_tests\_unit\background-tasks.test.ts:152:35)
    at Generator.next (anonymous)
    at asyncGeneratorStep (C:\Users\denni\Documents\CodingProjects\STEMM\node_modules\@babel/runtime\helpers\asyncGenerator.js:3:17)
    at _next (C:\Users\denni\Documents\CodingProjects\STEMM\node_modules\@babel/runtime\helpers\asyncGenerator.js:17:9)
    at C:\Users\denni\Documents\CodingProjects\STEMM\node_modules\@babel/runtime\helpers\asyncGenerator.js:22:7
    at new Promise (anonymous)
    at Object.<anonymous> (C:\Users\denni\Documents\CodingProjects\STEMM\node_modules\@babel/runtime\helpers\asyncGenerator.js:14:12)
    at Promise.then.completed (C:\Users\denni\Documents\CodingProjects\STEMM\node_modules\@babel/runtime\helpers\asyncGenerator.js:298:28)
    at new Promise (anonymous)
    at callAsyncCircusFn (C:\Users\denni\Documents\CodingProjects\STEMM\node_modules\jest-circus\build\utils.js:231:18)
    at callCircusTest (C:\Users\denni\Documents\CodingProjects\STEMM\node_modules\jest-circus\build\run.js:116:40)
    at runTest (C:\Users\denni\Documents\CodingProjects\STEMM\node_modules\jest-circus\build\run.js:252:11)
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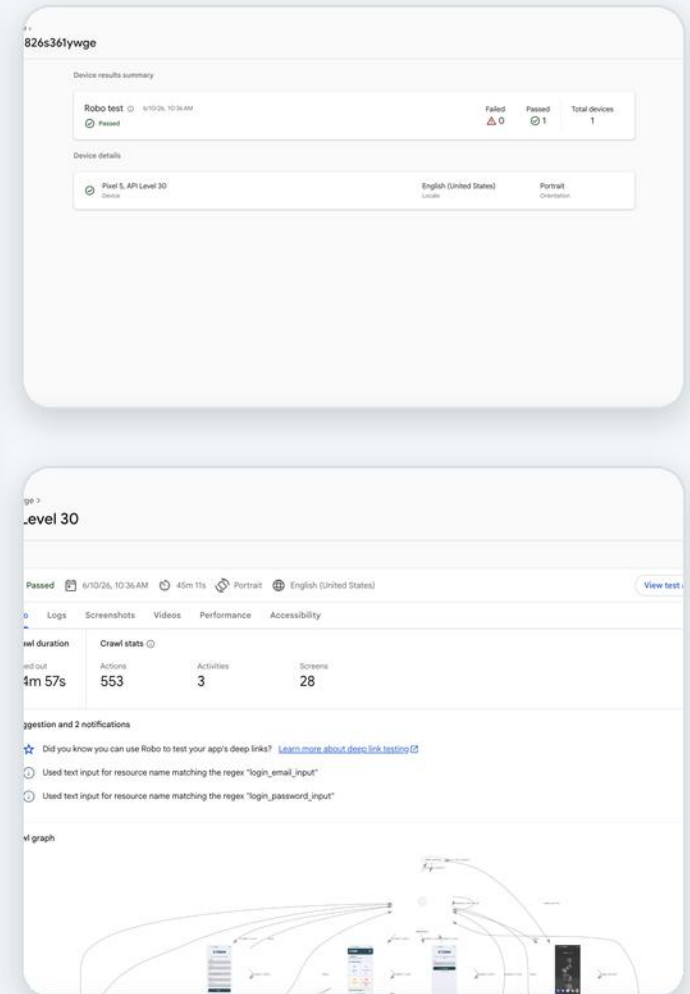
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at _throw (node_modules\@babel/runtime\helpers\asyncGenerator.js:20:9)

PASS _tests\_unit\crud.test.ts
PASS _tests\_e2e\app-config.test.ts
PASS _tests\_integration\Firebase.test.ts
PASS _tests\_unit\discriminant.test.ts
PASS _tests\_unit\points.test.ts
PASS _tests\_integration\theme.test.ts
PASS _tests\_unit\notifications.test.ts
# Console

console.warn
Notification permission not granted

39 |
40 | if (finalStatus !== "granted") {
> 41 |   console.warn("Notification permission not granted");
    |   ^
42 |   return null;
43 | }
44 |

at warn (lib/notifications.ts:41:13)
at asyncGeneratorStep (node_modules\@babel/runtime\helpers\asyncGenerator.js:3:17)
```



Device APIs need validation

Permissions, recording state, and sensor readings must be checked before allowing Next.

Firestore + SQLite split

Shared teams/leaderboard live in Firestore; local activity sessions/data points live in SQLite.

Duplicate scoring must be guarded

Client UI messages are not enough; point_awards prevents repeated team awards.

Test automation needs stable identifiers

Firebase Test Lab required login resource identifiers for email, password, and button actions.

Firestore

users / teams / point_awards / leaderboard

sync boundary

SQLite

challenge_sessions / data_points / ratings / GPS

1

Teacher dashboard

Class progress, team evidence, and activity completion overview.

2

Offline-first sync

Queue activity sessions locally, then sync when internet returns.

3

More activities

Expand beyond the current 7 STEMM challenges.

4

Better accessibility

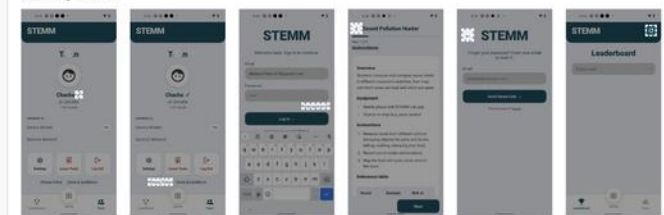
Use Firebase accessibility findings to improve target sizes, labels, and contrast.

5

Analytics dashboard

Show improvement over attempts and compare predictions vs outcomes.

Touch target size (8)



Firestore accessibility evidence

Why Firestore and SQLite?

Firestore is for shared team/leaderboard data. SQLite captures local activity sessions and data points.

How do points work?

Each completed activity gives 100 points. Duplicate completion for the same team/activity is blocked.

Why is this more than a quiz?

Students collect real evidence using sensors, video, predictions, measurements, and reflections.

Biggest challenge?

Making hardware APIs, validation, scoring, and automated testing work reliably together.



Performance evidence: startup and device metrics from Firebase Test Lab